

## Fine moduli spaces

Def Let  $M: \text{Sch} \rightarrow \text{Set}$  be a moduli functor.

If a scheme  $M$  represents  $M$ ,

i.e.  $\exists$  natural isomorphism  $\eta: M \rightarrow h_M$ ,

then we say  $M$  is a fine moduli space for  $M$ .

Rmk: ①  $\forall$  scheme  $S$ , have bijection:

$$\eta_S: M(S) := \{\text{families over } S\} / \sim_S \leftrightarrow h_M(S) := \{\text{morphisms } S \rightarrow M\}$$

Compatible with morphisms  $T \rightarrow S$

② Take  $S = \text{Spec } k$ , have

$$\{\text{objects}\} / \sim \leftrightarrow \{k\text{-points}\} := \{\text{morphisms } \text{Spec } k \rightarrow M\}$$

③ Take  $S = M$ , have

$$\eta_M^{-1}(\text{id}_M) \in M(M) \quad \leftarrow \text{id}_M \in h_M(M)$$

a (class of) family over  $M$

Def Let  $M$  be a fine moduli space for  $M$ ,  
we call  $\mathcal{U} := \eta_M^{-1}(\text{id}_M)$  the universal family.

Rmk: ① Let  $\tilde{F}$  be a family over  $S$ .

$$M(S) \xrightarrow{\eta_S} h_M(S)$$

$$f: S \rightarrow M$$

$$\tilde{F} \longmapsto f: S \rightarrow M \quad \leftarrow h_M(f): h_M(M) \rightarrow h_M(S)$$

$$f^* \mathcal{U} \longleftarrow \text{id}_M \circ f = h_M(f)(\text{id}_M)$$

$\therefore$  Every family can be obtained by pulling back  $\mathcal{U}$ .

② If  $(M, \eta)$  and  $(M', \eta')$  are fine moduli spaces for  $M$ ,  
 then there is a canonical isomorphism  $M \xrightarrow{\sim} M'$  (using  $\eta, \eta'$ )

Exercise 2.18 Skipped

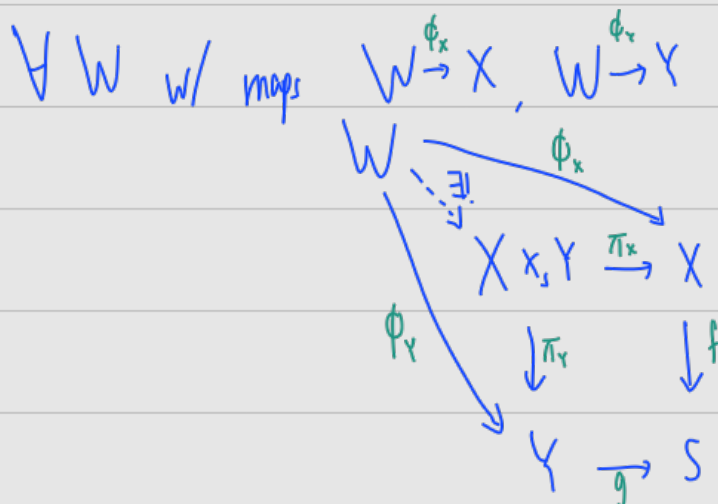
Will think of  $P^n$  as representing a moduli functor, but first, some more prerequisites:

★ Fiber product of schemes (over a base)

Fact (Hartshorne II, Thm 3.3)

Let  $S$  a scheme, along with morphisms  $X \xrightarrow{f} S$ ,  $Y \xrightarrow{g} S$ .

There is a fiber (fibred) product  $X \times_S Y$  w/ the universal property:  
 (with maps  $\pi_X: X \times_S Y \rightarrow X$   
 $\pi_Y: X \times_S Y \rightarrow Y$ )



In this reading group, write  $X \times Y$  short hand for  $X \times_{\text{Spec } k} Y$   
 (since we automatically have  $X \rightarrow \text{Spec } k$ ,  $Y \rightarrow \text{Spec } k$   
 by  $X, Y$  being  $k$ -schemes)

Rmk:  $\text{Spec } A \times_{\text{Spec } k} \text{Spec } B = \text{Spec } A \otimes_k B$  (by universal property of  $\otimes$ )

Eg.  $A^m \times A^n = \text{Spec } k[x_1, \dots, x_m] \otimes_k k[y_1, \dots, y_n] = \text{Spec } k[x_1, \dots, x_m, y_1, \dots, y_n] = A^{m+n}$

## \* Vector bundles (Vakil §4.1)

Def: Let  $X$  be a  $k$ -scheme, a rank  $n$  vector bundle is a map  $\pi: E \rightarrow X$  with

① each fiber  $\pi^{-1}(p) \cong A_k^n$

②  $\exists$  open cover  $\{U_i\}$  of  $X$  and isomorphisms  $\phi_i: U_i \times A_k^n \rightarrow \pi^{-1}(U_i)$   
 "trivializing open cover"  $\swarrow$   
 s.t. 
$$\begin{array}{ccc} U_i \times A_k^n & \xrightarrow{\phi_i} & \pi^{-1}(U_i) \subseteq E \\ \text{projection} \searrow & & \swarrow \pi|_{\pi^{-1}(U_i)} \\ & U_i & \end{array}$$
 commutes

If  $\{U_i\} = X$ , i.e.  $E \cong X \times A_k^n$ , we say that  $E$  is a trivial bundle.

A rank 1 vector bundle is called a line bundle.

(Example 2.20, also see Vakil §16.4 ( $Gr(d, n)$  over  $\text{Spec } \mathbb{Z}$  instead of  $\text{Spec } k$ )

Construction of  $Gr(d, n)$  as a moduli space

Fix a vector space  $V = A^n$ .

① Let  $Gr(d, n)$  be a moduli functor with

$$Gr(d, n)(S) = \{ \pi: \mathcal{E} \rightarrow S: \mathcal{E} \subseteq S \times V \text{ a rank } d \text{ subbundle} \} \quad (\text{no equivalence relation})$$

$$(?) \quad Gr(d, n)(f: T \rightarrow S) = [\pi \mapsto f^* \pi]$$

$$\begin{array}{ccc} T \times_S \mathcal{E} & \rightarrow & \mathcal{E} \\ f^* \pi \downarrow & & \downarrow \pi \\ T & \xrightarrow{f} & S \end{array}$$

② Want to represent  $\text{Gr}(d,n)$  by a scheme:

(Hoskin Exercise sheet 1 Q3)

Let  $\text{Gr}_{d,n}(k)$  be the set of  $d$ -dim subspaces of  $V = k^n$  (for  $k = \bar{k}$ )

(Following Hoskin's exercise 1, can glue together a scheme, or alternatively:  $\downarrow$ )

Fact: (Hoskin Exercise sheet 2 Q3, Vakil 16.4.D to 16.4.H)

We can treat  $\text{Gr}_{d,n}(k)$  as a closed subscheme of  $\mathbb{P}(\wedge^d k^n)$  by

$$\text{Gr}_{d,n}(k) \hookrightarrow \mathbb{P}(\wedge^d k^n)$$

$$\text{span}\{w_1, \dots, w_d\} \mapsto [w_1 \wedge \dots \wedge w_d]$$

③ Want to show  $\text{Gr}(d,n)$  represents  $\text{Gr}(d,n)$  (giving a fine moduli space)  
i.e. want to give natural isom.  $\text{Gr}(d,n) \rightarrow \text{H}_{\text{Gr}(d,n)}$ .

Let  $\mathcal{E} \subseteq V \times S$  a family over  $S$ .

Let  $\{U_\alpha\}$  be a trivializing open cover.

Then  $\forall \alpha$ , have  $U_\alpha \times A^d \hookrightarrow U_\alpha \times A^n$

$\parallel$

$\parallel$

$\mathcal{E}|_{U_\alpha}$

$U_\alpha \times V$

$\therefore$  Have  $U_\alpha \longrightarrow M_{n \times d}^d(k) := \{n \times d \text{ matrices of rank } d\}$

Composing:  $U_\alpha \longrightarrow M_{n \times d}^d(k) \longrightarrow \mathbb{P}(\wedge^d k^n)$  image lies in  $\text{Gr}(d,n) \subseteq \mathbb{P}(\wedge^d k^n)$

(Check!)

Gluing gives:  $f: S \longrightarrow \text{Gr}(d,n)$  (use transition functions)

$$\begin{array}{ccc} \therefore \text{We have} & \text{family} & \rightsquigarrow \text{morphism} \\ M(S) & \longrightarrow & h_{\text{Gr}(d,n)}(S) \\ E & \longmapsto & f \end{array}$$

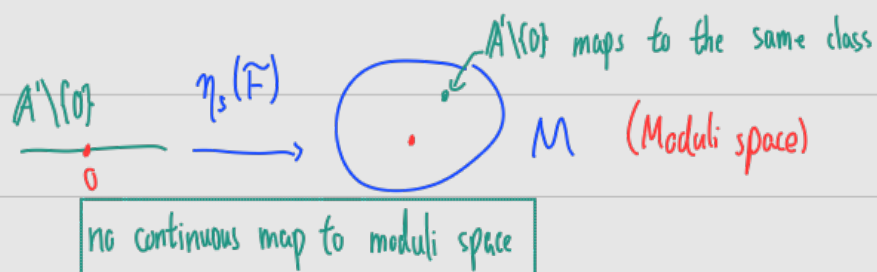
(Check!) This gives natural isom.

## Pathological behavior

A moduli functor will NOT have a fine moduli space of either:

① Jump phenomena: Have a family  $\tilde{F}$  over  $A'$  such that

$$\tilde{F}_s \sim \tilde{F}_i \text{ for all } s \in A' \setminus \{0\} \text{ but } \tilde{F}_0 \not\sim \tilde{F}_i.$$



② Unbounded: Cannot be represented by a finite type  $k$ -scheme

E.g. ①  $\text{End}_n$ : Endomorphisms of  $n$ -dim vector space.

Object:  $(V, T)$ :  $\dim_k V = n$ ,  $T \in \text{End}(V)$

Equivalence:  $(V, T) \sim (V', T')$  if  $\exists h: V \rightarrow V'$  s.t.

$$\begin{array}{ccc} V & \xrightarrow{h} & V' \\ T \downarrow \circlearrowright & & \downarrow T' \\ V & \xrightarrow{h} & V' \end{array}$$

S-Family:  $(\mathcal{F}, \phi): \mathcal{F} \rightarrow S$  rank  $n$  vector bundle,  $\phi \in \text{End}(\mathcal{F})$

Equivalence:  $(\mathcal{F}, \phi) \sim (\mathcal{G}, \gamma)$  if  $\exists h: \mathcal{F} \rightarrow \mathcal{G}$  s.t.

$$\mathcal{F} \xrightarrow{h} \mathcal{G}$$

$$\phi \downarrow \leadsto \downarrow \gamma$$

$$\mathcal{F} \xrightarrow{h} \mathcal{G}$$

$$\left[ \begin{array}{ccc} \mathcal{F} & \xrightarrow{\phi} & \mathcal{F} \\ & \downarrow \scriptstyle \phi & \downarrow \scriptstyle \phi \\ & S & \end{array} \right]$$

Let  $n=2$ ,  $(\mathcal{F}, \phi)$  family over  $A' = \text{Spec } k[s]$  with

$$\mathcal{F} = A' \times k^2 \quad (k^2 = A'^2)$$

$$\phi_s = \begin{pmatrix} 1 & s \\ 0 & 1 \end{pmatrix}$$

Then  $\mathcal{F}_s \sim \mathcal{F}_1$  for  $s \in A' \setminus \{0\}$ ,

but  $\mathcal{F}_1 \not\sim \mathcal{F}_0$ .

(By looking at  
Jordan forms)

Eg. ② Moduli problem: isom. classes of  $\mathbb{P}^1$ -vector bundles of rank 2, deg 0.

Suppose  $M$  is the moduli space for this problem (want contradiction)

Fact 1:  $[\mathcal{O}_{\mathbb{P}^1}(a) \oplus \mathcal{O}_{\mathbb{P}^1}(-a)] \in M$

Fact 2: For  $a \geq 1$ ,  $\dim H^0(\mathcal{O}(a) \oplus \mathcal{O}(-a)) = a+1$

Fact 3: Let  $M_n := \{[\mathcal{F}] \in M : \dim H^0(\mathbb{P}^1, \mathcal{F}) \geq n\}$ , then  $M_n \subseteq M$  closed.

$\therefore$  Have strictly decreasing chain

$$M \supsetneq M_2 \supsetneq M_3 \supsetneq \dots$$

$\dim M = \infty$  contradicting finite type.

## Coarse moduli space

Want to relax the notion of fine moduli space:  
Still want bijection of  $k$ -points, but less constraint on families.

Def A coarse moduli space for a moduli functor  $M$  is a scheme  $M$

and a natural transformation  $\eta: M \rightarrow h_M$  s.t.:

(a)  $\eta_{\text{Spec } k}: M(\text{Spec } k) \rightarrow h_M(\text{Spec } k)$  is bijective

(b) For any scheme  $N$  and natural transformation  $\nu: M \rightarrow h_N$ ,  
 $\exists!$  morphism of schemes  $f: M \rightarrow N$  s.t.  $\nu = h_f \circ \eta$ .

where  $h_f: h_M \rightarrow h_N$  is given by

$$\text{Hom}(-, M) \rightarrow \text{Hom}(-, N)$$

$$\phi \mapsto f \circ \phi$$

$$\begin{array}{ccc} M & \xrightarrow{\eta} & h_M \\ & \searrow \nu & \downarrow h_f \\ & & h_N \end{array} \quad \begin{array}{c} \exists! \\ \downarrow \end{array}$$

Rmk-Prop: A fine moduli space is a coarse moduli space

Pf: Suppose  $(M, \eta)$  is a fine moduli space,

Then we have  $\eta^{-1}: h_M \rightarrow M$ .

Let  $\nu: M \rightarrow h_N$ .

$$\begin{array}{ccc} M & \xrightleftharpoons{\eta^{-1}} & h_M \\ & \searrow \nu & \downarrow \\ & & h_N \end{array}$$

Then take  $f$  s.t.  $h_f = \nu \circ \eta^{-1} \Rightarrow h_f \circ \eta = \nu$

$f$  exists by Yoneda: Take  $f = (\nu \circ \eta^{-1})_M(\text{id}_M): M \rightarrow N$

Rmk: Coarse moduli space  $M$  is unique up to unique isom.:

If  $(M, \eta)$ ,  $(M', \eta')$  are coarse moduli spaces for  $M$ ,  
can construct canonical isom (using  $\eta, \eta'$  and Yoneda)

Prop-Exercise 2.26 Let  $(M, \eta)$  be a coarse moduli space for  $M$ .

Then  $(M, \eta)$  is a fine moduli space iff

(1)  $\exists$  a family  $\mathcal{U}$  over  $M$  s.t.  $\eta_M(\mathcal{U}) = \text{id}_M$ , and

(2) For families  $\tilde{F}$  and  $\tilde{G}$  over  $S$ , we have  $\tilde{F} \sim_s \tilde{G} \Leftrightarrow \eta_S(\tilde{F}) = \eta_S(\tilde{G})$ .

Lemma 2.27 Jump phenomenon  $\Rightarrow$  no coarse moduli space.

Pf: Let  $\tilde{F}$  be an  $A'$  family w/  $\tilde{F}_s \sim \tilde{F}$  for  $s \neq 0$  and  $\tilde{F}_1 \not\sim \tilde{F}_0$ .  
Have  $\eta_{A'}(\tilde{F}) =: f: A' \rightarrow M$ .

Then for a  $k$ -point  $s: \text{Spec } k \rightarrow A'$ , have  $f \circ s = \eta_{\text{Spec } k}(\tilde{F}_s)$ ,

and for  $s \neq 0$ ,  $\tilde{F}_s = \tilde{F} \Rightarrow f \circ s = f \circ 1 =: m: \text{Spec } k \rightarrow M$ .

$\Rightarrow f$  is constant on  $A' \setminus \{0\}$

pre image  $f^{-1}(m)$   
is closed  $\Rightarrow f$  is constant on  $A'$

$\Rightarrow f \circ 0 = f \circ 1 = m$  But  $\tilde{F}_1 \not\sim \tilde{F}_0$   
 $\downarrow \downarrow$   
 $m$

contradiction:  $\eta_{\text{Spec } k}$  not a bijection.

## Construction of moduli spaces:

Step ①: Fix a discrete invariant (which is invariant under equiv. relation)

e.g. • For  $Gr(d,n)$  fixing dimension  $d$

• For vector bundles, fix rank and degree

• For curves, fix genus

Step ②: Restrict to a reasonable class of objects which are "bounded".

Usually: "stable" objects.

Step ③: Find a family  $\tilde{F}$  over a scheme  $P$  with the local univ. property (see below)

We call  $P$  a parameter space, as  $k$ -points of  $P$  surjects onto  $A/\sim$ .

Typically  $P(k) \rightarrow A/\sim$  is not a bijection.

( $P$ : Space before modding out relation)

Step ④: Find a group  $G$  on  $P$  s.t.  $G.p = G.q \Leftrightarrow \tilde{F}_p \sim \tilde{F}_q$ .

Then  $P(k)/G \cong A/\sim$

(Main focus of the coming few weeks: Groups and quotients)

Step ⑤: Usually the  $G$ -action is algebraic. Taking scheme quotients (§3-5) will give moduli space. Will use Mumford's Geometric Invariant Theory (GIT)

Def For a moduli problem  $\mathcal{M}$ , a family  $\tilde{F}$  over  $S$  has the local universal property if  $\forall$  family  $\mathcal{G}$  over  $T$  and  $\forall t \in T(k)$ ,  
 $\exists$  open nbd  $U \subseteq T$  and a morphism  $f: U \rightarrow S$  s.t.  $\mathcal{G}|_U \sim_U f^* \tilde{F}$ .  
(not necessarily unique)

Compare to: Universal family  $\mathcal{U}$  on  $M$ :

$\forall$  family  $\mathcal{G}$  over  $T$ ,  $\exists!$  morphism  $f: T \rightarrow M$  s.t.  $\mathcal{G} \sim_T f^* \mathcal{U}$ .